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An Evaluation of a Major Expansion in Automated License Plate Reader (ALPR) Technology

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ABSTRACT

Automated license plate readers (ALPRs) are one of the most recent technological advancements that have rapidly diffused across US law enforcement. However, empirical research on ALPRs has not kept pace with their diffusion as there have only been a few rigorous evaluations of the technology's effectiveness in addressing crime. This study builds on the scant research base by performing both a process- and impact-evaluation of a major ALPR technology expansion in the Atlantic City (NJ) Police Department (ACPD), which is located on a barrier island. A Smart Policing Initiative (SPI) grant allowed the department to buy and install stationary/fixed ALPRs on all entrances and exits to scan every moving vehicle in and out of the city's jurisdiction. We explored the project's challenges and implementation fidelity, plus the technology's impact on investigations and clearance rates for shooting incidents as well as whether the technology expansion was associated with crime reduction. Descriptive statistics, bivariate associations and interrupted time-series analyses found that use of the technology may be more siloed in certain investigative units rather than for patrol functions. While the ALPR expansion did not reduce violent crime, it was associated with reductions in shootings, motor vehicle thefts, and property crime.

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Police; technology; ALPRs; process evaluation; impact evaluation; implementation science

A number of technological innovations have rapidly diffused across the law enforcement profession in the past twenty years. They include officer-body worn cameras (BWC), facial recognition, acoustic gunshot detection systems (AGDS), and automated license plate readers (ALPR). While many rigorous evaluations of BWCs have been conducted, thanks in part to generous funding streams since 2015 (Bureau of Justice Assistance, n.d.), empirical research on the other innovations has not kept pace with their diffusion (Johnson et al., 2022; Mares, 2022). This diffusion-to-evaluation ratio is particularly poor for ALPRs: cameras that scan the alphanumeric license plates on moving or parked vehicles and compare them to existing databases, such as lists of

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stolen cars or registered owners who have open warrants. By 2009, over one-third of all police agencies in the US with more than 100 sworn officers had acquired ALPRs (Lum et al., 2010) with 64.9% of all municipal departments with 100+ officers using the technology as of 2020 (Law Enforcement Management and Administrative Statistics [LEMAS], 2020). Yet, the extant literature has featured only a handful of evaluations testing the effectiveness of ALPRs on both patrol functions (Kernahan & Valasik, 2019; Lum et al., 2011; Taylor et al., 2012; Wheeler & Phillips, 2018) and investigations (Koper & Lum, 2019).

Research indicates ALPRs provide increased efficiency in terms of generating more “alerts” for stolen plates or vehicles (Potts, 2018; Roberts & Casanova, 2012), which translate into arrests and automobile recoveries for such crimes being two-to-three times more likely during patrols when the technology is used (Koper et al., 2019). Thus far, however, there is little evidence that ALPRs reduce crime and even less is known about their impact on solve/clearance rates (see Koper & Lum, 2019 for an exception). Although many of the aforementioned studies included rigorous designs with pre- and post-intervention periods as well as treatment and control areas, the modal evaluation involved either a single or a few ALPRs mounted on police vehicles with officers using them to patrol or in conjunction with roadblocks/checkpoints in hot spot locations (Lum et al., 2011; Taylor et al., 2012; Wheeler & Phillips, 2018). Koper et al. (2019), in fact, discuss how these place-based functions might limit the technology’s utility, and they suggest that patrolling larger geographic areas may be more optimal for mobile ALPRs. Additionally, only a single evaluation has examined the impact of stationary/fixed ALPRs mounted on traffic lights, light posts, etc. (Kernahan & Valasik, 2019).

This paper builds on the extant literature by performing a process and outcome evaluation of a major expansion of ALPR technology in the Atlantic City Police Department (ACPD). Atlantic City, New Jersey is located on a barrier island with limited access points. While ACPD previously operated a handful of mobile ALPRs, a Smart Policing Initiative (SPI) grant allowed the department to buy and install stationary/fixed ALPRs on all entrances and exits to scan every moving vehicle in and out of the city’s jurisdiction. We not only explore the project’s challenges and implementation fidelity, but also examine the technology’s impact on crime—including interrupted time series analyses—as well as use in investigations and clearance rates for gun violence.

ALPRs: What We Know

Acquisition, Prevalence, and Usage

ALPR acquisition and usage in the US varies by size, agency type, and strategic goals. Prevalence increases with agency size (Strom, 2017), and much of the growth in rates of usage among large departments (i.e., >100 sworn officers) occurred between 2009 and 2016 (LEMAS, 2016). They are more common among local/municipal departments, followed by county/sheriff’s offices, and state police/highway patrol. Among all types of agencies with more than 1,000 sworn officers, the vast majority –84 to 94%—use the technology (LEMAS, 2020). Agencies were more likely to acquire ALPRs when

placing a higher strategic value on predictive policing as well as those that valued both community- and hot spots-policing among a subsample of departments with 250+ officers (Hendrix et al., 2019), although the processes by which departments decide to acquire ALPRs are not well known (see also Strom, 2017).

There are two ways to deploy ALPRs: mobile units attached to police vehicles or stationary/fixed units that are mounted on utility poles, traffic lights, billboards, bridges, etc. They can also be placed on portable mobile surveillance trailers (Mares & Maier, 2022). Mobile units allow for versatile placement that is adaptable depending on strategic goals (e.g., large events), yet they are dependent on the vehicles they are mounted on, which can be taken out of service due to maintenance and other issues (Gierlack et al., 2014). Stationary/fixed units cover extremely limited geographic space—usually one lane in one direction on a single road per camera—but scan every single vehicle that passes when operational. They also require electricians for installation and upkeep—not to mention access must be granted by the owner/authority of the property that the ALPR is being mounted on. The majority of deployed ALPRs are mobile units mounted on police vehicles (Lum et al., 2019). Some agencies report using both mobile and stationary/fixed units (Roberts & Casanova, 2012; Willis et al., 2018). Most departments that use the technology, however, typically deploy a small number of ALPRs. Lum et al. (2019)'s nationally representative sample of agencies with 100+ officers, which was conducted in 2014, found that 75% using the technology owned 7 or fewer cameras and 90% owned fewer than 15 cameras; 5% of agencies owned more than 25 ALPRs.

Effectiveness: Patrol and Investigatory Functions

ALPR evaluations that have tested the technology's effectiveness on crime reduction are limited to just a few jurisdictions. ALPRs did not significantly reduce crime in treatment areas or post-intervention periods in Mesa, AZ (Taylor et al., 2012), Alexandria City and Fairfax County, VA (Lum et al., 2011), and "Tigerland"—the entertainment district at Louisiana State University in Baton Rouge, LA (Kernahan & Valasik, 2019). Alternatively, Wheeler and Phillips (2018) found that ALPRs used in concert with checkpoints/roadblocks in Buffalo, NY decreased both part 1 violent crimes and traffic accidents with the caveat that findings were sensitive to the modeling strategies employed. The evaluations in Mesa, AZ; Alexandria City and Fairfax County, VA; and Buffalo, NY deployed mobile ALPRs mounted on vehicles, whereas the Baton Rouge, LA evaluation relied on stationary/fixed ALPRs in Tigerland. It is unclear whether the lack of crime reduction effects is due to the limited patrol coverage in hot spot areas (compared to city-wide) (see Koper et al., 2019), the small number of ALPRs typically deployed (Lum et al., 2019), low base rates of crimes in general, and/or the possibility that offenders are unaware that the technology is being used.

It is also unclear what the mechanism(s) at work are that could potentially lead to reductions in crime (see Pawson & Tilley, 1997). If the general public were made aware of the ALPRs—either through signage, press releases, media coverage, and/or word of mouth—then perhaps there could be a general deterrent effect. Aside from patrol-related functions, ALPRs may assist law enforcement with investigative and intelligence-related efforts, which can have a specific deterrent or incapacitation effect

if suspected offenders are identified, arrested, and incarcerated. This research base is even more limited. Only one study, to our knowledge, examined whether clearance rates for auto theft and robbery improved after the expansion of a stationary/fixed ALPR network. While case clearances for these offense types did improve after expansion, these changes were not statistically significant and the data point to other relevant factors—such as the use of other technologies (e.g., real-time crime center; non-ALPR surveillance cameras) and changes in police management—that may have contributed to the increases (Koper & Lum, 2019).

Some agencies do recognize that ALPRs can act as a force multiplier: adding to existing information technology systems that collect vital data that can be stored, shared, analyzed, searched, and retrieved (Byrne & Marx, 2011). Willis et al. (2018) conducted focus groups with sworn and civilian staff (e.g., crime analysts) in one department that reported on the novel ways they were using ALPR technology, including identifying suspect movements and confirming alibis. This particular agency, however, had invested in a large network of both mobile and stationary/fixed ALPRs located throughout its jurisdiction, which fostered such investigatory and intelligence-related efforts. The available evidence, albeit dated, suggests that most agencies will not be in a position to leverage data collected by ALPRs into their full investigative potential. As of 2014, the majority of police departments with ALPR technology have fewer than 7 cameras (Lum et al., 2019). That same survey found that only 10–12% of agencies with ALPRs “always or almost always” use the technology for investigating crimes against persons or property.

Citizens’ and Officers’ Perceptions of ALPRs

A number of citizen and legal/activist groups, including the American Civil Liberties Union (2013), have raised privacy concerns regarding ALPRs (see also California State Auditor, 2020). There is a degree of support that the public may view some of the technology’s functions with skepticism, but published studies have only relied on a single survey of community residents in Fairfax County, VA in 2010. While most respondents were unfamiliar with ALPRs, they were generally supportive of the technology, especially to detect stolen vehicles and scan whether registered owners were wanted or had outstanding warrants; there was less support for ALPRs to scan vehicles for unpaid parking violations and other citations (Merola et al., 2014). Citizens with favorable views of police and those who identified the data being collected as public information were more likely to support ALPR use, while racial/ethnic minorities (Merola and Lum, 2014) as well as those provided with an experimental condition of increased knowledge of ALPRs (Merola et al., 2019) were less supportive.

Only one published study, to our knowledge, has surveyed police officers’ perceptions of ALPRs. Shjarback (2024) examined these perceptions among officers in a single agency that had previously operated a few mobile ALPRs prior to a major expansion of the technology. Those with prior experience using the technology, younger and more inexperienced officers, and stronger guardian perceptions viewed the ALPR expansion more favorably. Prior experience using ALPRs also served as a protective factor: lowering officers’ level of skepticism toward the technology expansion. Overall, relatively little is known about how both police officers and citizens view ALPRs.

Current Study

Although ALPRs offer increased efficiency and automate a process that was previously conducted manually by police officers, the available evaluative research has not found much support that the technology improves effectiveness in the form of reduced crime. These findings are derived from a small number of research sites and agencies that implemented ALPRs on a limited basis: placed-based tactics in smaller geographic areas (i.e., hot spots). Moreover, most of the evaluations—and police departments that use the technology more broadly—employed low numbers of ALPRs that prohibit them from meaningfully contributing to robust information systems that can be used for investigative and intelligence-related purposes. Despite many threats to internal validity ruled out, it is still unclear what the potential mechanism(s) could be.

Atlantic City's ALPR expansion program and its subsequent evaluation is unique. The city's jurisdiction is located on a barrier island on the South Jersey coast of the Atlantic Ocean with only 8 different routes in and out. Such a geographic location makes it feasible to place stationary/fixed ALPRs on every entrance and exit to scan all vehicles traveling to and from the city, which is more cost-prohibitive in most other jurisdictions. Prior to applying for and receiving grant funding in 2020 from the Bureau of Justice Assistance's (BJA) SPI to purchase and install stationary/fixed ALPRs at all entrances and exits in and out of Atlantic City, ACPD documented notable year-over-year increases in both motor vehicle thefts (20%) and firearm assaults (36%). In addition, ACPD had reason to believe that the two crime types were related. Reports of stolen vehicles often occurred within 24-h of a shooting, and several long-term investigations by ACPD's Violent Crimes Unit (VCU) showed that gangs used juveniles to steal vehicles that were later used either during drive-by shootings or to transport involved persons to and from the crime scene. Vehicles were involved in 53% of shootings in a nearly two-year period immediately before the evaluation began. At the time, transactional gun violence was occurring on a sustained basis between two rival Atlantic City-based criminal street gangs—one of which was allied with another gang in an adjacent jurisdiction. Shooting incidents between members and associates in this triangle frequently occurred in Atlantic City and several neighboring jurisdictions, including Pleasantville and Egg Harbor.

ACPD believed stationary/fixed ALPRs at all entrances and exits that scan every vehicle's plate would enable law enforcement to detect stolen cars coming in and out of Atlantic City either before or after shooting incidents. It was thought that alerts for stolen vehicles as well as other violations would allow ACPD to engage in patrol-related functions (i.e., vehicle stops and interdictions) and investigation-related activities, such as identifying suspects and travel patterns, gathering evidence and building cases, and clearing cases through arrest. As such, we performed both a process and outcome/impact evaluation of the ALPR technology's expansion.

Methods

Research Site and Data

Atlantic City is 11 square miles located on Absecon Island—a barrier island (i.e., land separated from the mainland by shallow water)—off the Atlantic coast of southern New Jersey. As of the 2020 census, the city's local resident population was 38,497;

however, it is a popular tourist destination known for its boardwalk, beaches, casinos, and events (e.g., concerts, comedy shows, conferences). Atlantic City, therefore, hosts approximately 30 million visitors annually, making it the most visited destination in the state and the 25th most visited tourist destination in the US. The ACPD employs approximately 250 sworn, full-time officers.

Despite these levels of tourism, Atlantic City has a high degree of economic and social disadvantage compared to the state as a whole. According to the 2022 American Community Survey's 5-year estimates, median household income is \$35,188 (NJ = \$99,781), 32.4% of residents live in poverty (NJ = 9.7%) including 46.9% of children under 18 years old, 19.2% possess a Bachelor's degree or higher (NJ = 43.8%), and 30.8% own their homes (NJ = 63.7%), (US Census Bureau, n.d.). In terms of racial/ethnic demographics from the 2020 Census, 35.6% of local residents were Black, 29.9% were Hispanic; 16.2% were white, non-Hispanic, 16.2% were Asian, and 17.1% were classified as some other race—making it a minority-majority city (US Census Bureau, n.d.). Associated with these levels of disadvantage, Atlantic City's crime rate, including violence and gun crime, is relatively high compared to both New Jersey and the US. For example, from 2019 to 2021, Atlantic City had the 7th highest rate of fatal and injurious shooting victims (384 per 100,000 residents) among all municipalities in New Jersey compared to the state average of 40 per 100,000 residents (Shjarback, 2023).

Given that Atlantic City is located on a barrier island, there are only 8 ways in and out of the jurisdiction: the Atlantic City Expressway (ACE), Route 30, Route 40, the Brigantine Bridge to the north, and four roads that connect Atlantic City to Ventnor City in the south (West End Avenue, Winchester Avenue, Ventnor Avenue, and Atlantic Avenue). BJA's SPI grant allowed for stationary/fixed ALPRs to be installed on all lanes and in both directions of these 8 roadways. Figure 1 shows Atlantic City along with its neighboring jurisdictions as well as the locations of each of the ALPRs. Data from the New Jersey Department of Transportation estimate that over 193,000 motorists travel these roadways daily—primarily the ACE, Route 30, and Route 40 to and from the mainland—with over 70 million annually. Theoretically, every route to and from Atlantic City would be monitored and all vehicles would be subjected to not only license plate scans but a picture and record of each plate with a geographic camera location, date, and time.

The data were generated from the ACPD, specifically the VCU and the agency's two crime analysts, as well as the ALPR vendor who provides information to the ACPD on the number of active hours each camera is scanning, total scans, total alerts, and a breakdown of the type of alert (e.g., stolen vehicle, suspended or revoked).

Measures

The variables were broken down into process- and outcome/impact-based measures. All process-based measures focused on how the ALPRs performed and what the ACPD and its officers did, per se. They included the total number of active hours that the stationary/fixed ALPRs were scanning, total licenses plates scanned, total alerts, and type of alert as well as a breakdown by inbound and outbound functions and each camera location. Additionally, the process-based measures include the number of



Figure 1. Atlantic city, neighboring jurisdictions, and ALPR locations.

Note: "X"s denote approximate ALPR cameras locations.

vehicle stops initiated following ALPR alerts and the results of those stops, such as arrests made and stolen vehicles or plates recovered.

The outcome/impact-based measures, by contrast, were related to examining whether the ALPR technology has assisted in reducing crimes and aiding with investigations. They include monthly Part 1 equivalent National Incident-Based Reports System (NIBRS) offenses for violent and property crimes as well as specific focuses on shootings and motor vehicle thefts. The frequency with which detectives used data from the ALPR information system for investigative purposes in shooting incidents as well as arrest/clearance rates for shooting incidents were also explored.

Analytic Strategy

The analysis proceeded in three broad phases. First, we present a set of descriptive/summary statistics that focus on the "process" evaluation of what the ALPR cameras

did and how they operated as well as how ACPD and its officers responded to ALPR alerts. We also discuss the logistical challenges that the department encountered and some of the impediments to the installation and successful operationalization of the technology. Next, we explored if investigatory functions changed from pre- to post-ALPR technology expansion, specifically VCU detectives' use of the system for shootings/homicide cases and arrest/clearance rates. The same bivariate relationships were also assessed between monthly offenses for violent crime, property crime, shootings, and motor vehicle thefts from pre- to post-expansion. Lastly, an interrupted time-series analysis (e.g., ARIMA) tested the impact that the ALPR expansion, as an intervention, had on monthly crime offenses. Given the jurisdiction-wide implementation of the stationary/fixed ALPR expansion—rather than more specific, place-based interventions that allow for treatment and comparison groups—we were precluded from conducting a more rigorous evaluation with a higher degree of statistical controls. We were unable to compare ACPD's crime trends with equivalent data from similarly-situated jurisdictions in the state due to data limitations.¹

Findings

Descriptive/Summary Statistics

ACPD experienced substantial delays from their proposed timeline installing and making the ALPR cameras operational. The purchasing and procurement of the cameras ran smoothly; however, delays occurred due to issues with the camera installation. Given that the ALPRs were to be mounted on stationary infrastructure, such as utility poles, traffic lights, and overpasses, locations needed to be strategically picked to allow for unobstructed lines of sight for the cameras to properly read each passing vehicle's license plate. All registered motor vehicles in New Jersey must display both front and rear plates; however, this is not mandatory in other states that only require rear plates, particularly neighboring Pennsylvania. All cameras, therefore, needed to be placed to face and scan each vehicle's rear license plate in both inbound and outbound directions. Moreover, not all roads and subsequent infrastructure that were ideal locations for ALPR placement were owned by the city of Atlantic City. Some were either county roads or owned by the state or a separate entity, which required permission and sometimes formal agreements. The ACE for example, which is one of the most traveled routes in and out of the city—especially among non-locals, falls under the jurisdiction of the South Jersey Transportation Authority (SJTA). The memorandum of agreement (MOA) with the SJTA took longer than expected and delayed ALPR camera installation on the ACE. Lastly, stationary/fixed ALPR installation requires

¹The New Jersey State Police (NJSP) once compiled and provided monthly UCR data for all municipal police departments in the state, which would have offered equivalent temporal measures for a number of similarly-situated jurisdictions that had not invested in a major ALPR expansion. However, NJSP moved to quarterly crime measures in recent years and those measures have not been updated since Q2 of 2023. As a result, we tried to gather measures of monthly crime from other data sources, including AH Analytics' "Real-Time Crime Index" (RTCI). Yet, RTCI only includes two agencies in New Jersey: Newark (the state's largest city of approx. 304,000) and North Bergen.

coordination and collaboration with electric/utilities companies for mounting, powering, and servicing the technology, which can postpone or interrupt operations.

With the exception of the ACE, all ALPRs were installed and powered in late November and early December of 2022. ACPD began live monitoring in mid-to-late December after officers and detectives participated in in-service training on how to use the ALPR system and reviewed both the department’s and the New Jersey Attorney General’s ALPR policies. ACPD has mentioned the ALPR expansion project in the press; however, it has not received much attention or coverage by and large. The location of each camera, direction—inbound versus outbound—and the monthly active hours of operation from November 2022 through November 2024 are presented in [Tables 1](#) and [2](#), respectively. White denotes cameras that were largely operational for more than 600h during the month, gray denotes cameras that were partially operational with between 1 and 599h during the month, and black denotes cameras that were inactive with zero hours of scanning during the month. Upon an early examination of each camera’s active hours of operation, total scans, and alerts, results uncovered that a number of ALPRs either needed to be repowered, reset, or their camera angles readjusted since license plates were not being properly read and documented in the information system. The cameras in both directions on the Brigantine Bridge did not initialize and were replaced; both the ALPR vendor and electric/utilities company needed to make several site visits in January and February of 2023 and beyond for maintenance.

Total Active Hours, Scans, and Alerts

From November 2022 through November 2024, the stationary/fixed ALPRs were active for 234,125 h. However, if all twenty-one cameras in both directions were fully operational 24 h a day/7 days a week, then they would have scanned for 383,544 total hours across the aforementioned time period. Because some ALPR cameras only

Table 1. ALPR cameras inbound: Active hours by month.

	22	22	23	23	23	23	23	23	23	23	23	23	23	23	24	24	24	24	24	24	24	24	24	24	24
	11	12	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8	9	10	11
ACE Lane 1																									
ACE Lane 2																									
ACE Lane 3																									
Route 30																									
Route 40																									
Lane 1																									
Brigantine Bridge																									
Atlantic/Ohio																									
Ventnor/Jackson																									
West End Ave																									
Winchester Ave																									

Note: White > 600 active hours; Gray < 600=active hours; Black = 0 active hours

Table 2. ALPR cameras outbound: Active hours by month.

	22	22	23	23	23	23	23	23	23	23	23	23	23	23	24	24	24	24	24	24	24	24	24	24	24
	11	12	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8	9	10	11
ACE Lane 1																									
ACE Lane 2																									
ACE Lane 3																									
Route 30																									
Route 40																									
Lane 1																									
Lane 2																									
Brigantine Bridge																									
Atlantic/Jackson																									
Ventnor/Jackson																									
West End Ave																									
Winchester Ave																									

Note: White > 600 active hours; Gray < 600=active hours; Black = 0 active hours

worked partially during certain months and others did not work at all, the stationary/fixed ALPR system provided 61.0% of active hours of coverage. Tables 1 and 2 highlight how it took some time before many of the cameras became largely operational before improving. During the first five months from November 2022 through March 2023, only 26 out of 105 camera location-months scanned for more than 600h per month (24.8%), another 29 scanned for less than 600h per month (i.e., partial; 27.6%), and nearly half ($n=50$; 47.6%) were inactive with 0h of reads per month. During the active hours for the entire twenty-five-month time period, these ALPRs documented 78,859,548 total reads/scans of license plates and 1,978,578 alerts.

In terms of the breakdown of the total alerts, the vast majority were for either “other” ($n=1,752,299$; 88.6%) or the registered owner’s license being “suspended or revoked” ($n=181,984$; 9.2%). All other types of alerts occurred much less frequently, including for “stolen plate” ($n=20,305$; 1.0%), “stolen vehicle” ($n=12,384$; 0.6%), “unknown” ($n=7,224$; 0.4%), “sex offender” ($n=2,290$; 0.1%), “wanted person” ($n=1,663$; 0.08%), and “violent gang” ($n=294$; 0.01%). Alerts for the first two categories occurred continuously: with “other” averaging 2,302.6 per 24h (95.9 per hour) and “suspended or revoked” averaging 239.1 per 24h (10.0 per hour). Relatively more serious “stolen plate” and “stolen vehicle” alerts are more manageable: occurring, on average, 26.7 times and 16.3 times per 24h, respectively. Still, this means ACPD is being alerted to a stolen plate about once per hour and a stolen vehicle approximately once every less than two hours for cameras scanning in both directions—inbound and outbound.

Inbound Cameras

Inbound scans and alerts are more amenable to ACPD officers’ action and output since the vehicle in question is coming into their jurisdiction. 32,304,725 license plates

were read/scanned inbound from November 2022 through November 2024—making up 41.0% of the total reads/scans by the stationary/fixed ALPRs during this time period. The 41.0% is due to the fact that the inbound cameras did not perform as well as the outbound cameras. Out of the 250 camera-months (i.e., 25 months X 10 separate cameras), the inbound ALPRs were largely operational 44.0% of the time, partially operational 23.2% of the time, and inactive 32.8% of the time compared to the outbound ALPRs, which were largely operational 60.4% of the time, partially operational 22.2%, and inactive only 17.5% of the 275 camera-months (25 months X 11 separate cameras).

Of the 32,304,725 reads/scans, there were 828,220 alarms from inbound cameras (41.9% of all the total stationary/fixed ALPR alarms). Following the broader trends, the vast majority of inbound alerts were for either “other” ($n=717,948$; 86.7%) or the registered owner’s license being “suspended or revoked” ($n=74,512$; 9.0%). All other types of alerts occurred much less frequently, including for “stolen plate” ($n=6,613$; 0.8%), “stolen vehicle” ($n=4,428$; 0.5%), “unknown” ($n=2,806$; 0.3%), “sex offender” ($n=801$; 0.1%), “wanted person” ($n=625$ 0.08%), and “violent gang” ($n=120$; 0.01%). While not occurring as frequently as cameras scanning in both directions, inbound alerts averaged 943.4 per 24 h for “other” (39.3 per hour), 97.9 per 24 h for “suspended or revoked” (4.1 per hour), 8.7 per 24 h for “stolen plate” and 5.8 per 24 h for “stolen vehicle.” ACPD was alerted to a stolen plate coming into the city about once every three hours and an inbound stolen vehicle approximately once every five hours. These rates, importantly, are based on the total 18,264 h between November 2022 and November 2024. They would be substantially higher had the cameras’ active hours increased.

Outbound Cameras

During the 25-month time period, 46,554,823 license plates were read/scanned by outbound cameras (59.0% of the total stationary/fixed ALPR reads). Of those reads/scans, ACPD was alerted 1,150,362 times (58.1% of stationary/fixed alerts) with a breakdown of 1,034,351 for “other” (89.9%), 107,472 for a registered owner’s license being “suspended or revoked” (9.3%), 13,692 for a “stolen plate” (1.2%), 7,956 for a “stolen vehicle” (0.7%), 4,418 “unknown” (0.4%), 1,489 for a “sex offender” (0.1%), 1,038 for a “wanted person” (0.09%), and 174 times for “violent gang” (0.02%). Outbound alerts averaged 1,359.2 per 24 h for “other” (56.6 per hour), 141.2 per 24 h for “suspended or revoked” (5.9 per hour), 18.0 per 24 h for a “stolen plate” and 10.5 per 24 h for a “stolen vehicle.” ACPD was alerted to a stolen plate leaving the city about once every forty-five minutes and an outbound stolen vehicle approximately once every hour and thirty minutes.

One point about the alerts: the ALPR system scans and searches existing databases using the alpha numeric license plate, but it cannot verify the state. Therefore, hypothetically, if a stolen vehicle with a New Jersey license plate of “ABC123” is reported stolen and a Pennsylvania license plate of “ABC123” is scanned, then the system will hit on and issue a false alert. The ALPRs also scan non-license plates on occasion. For example, lettering on the side of a truck might get scanned, which may produce other types of false alarms.

ACPD Responses to ALPR Alerts

Based on the number of alerts from stationary/fixed ALPR cameras, it appears as though ACPD is understaffed and unable to respond to such a staggering volume. There were only 87 documented patrol-related responses (as opposed to investigatory functions) to active ALPR alerts from November 2022 through December 2024. Of those 87 alerts, 52 were for stolen vehicles (59.8%), 13 for stolen plates (14.9%), 14 for wanted persons (i.e., warrants) (16.1%), and 8 for suspended or revoked license/registration (9.2%). Seventy-two vehicle stops were initiated with 40 stolen vehicles and 9 stolen license plates recovered from those initiated stops. Additionally, the vast majority of these 72 stops resulted in individuals being charged and taken into custody ($n=66$) or issued a summons ($n=1$). On other occasions, 18 stolen vehicles and 5 stolen license plates were recovered after being found abandoned/unoccupied for a total of 58 recovered stolen vehicles and 14 recovered stolen licenses plates over the 26-month period. It was difficult for the ACPD to synthesize and arrive at accurate numbers of recovered stolen vehicles or plates prior to November 2022. Aside from largely inbound-related ALPR alerts, ACPD provided 59 notifications to neighboring police departments regarding outbound ALPR alerts. Given the fact that the ALPR expansion was not widely covered—despite ACPD not hiding it—and there was a small workload impact on patrol-related functions, it appears that any potential crime reduction benefits would not be yielded from a general deterrent effect.

Bivariate Relationships Examining Pre- and Post-ALPR Expansion

In addition to the descriptive statistics that explain some of the process-based measures, we attempted to examine whether the ALPR expansion was associated with any changes in the use of the technology for investigative purposes. Given the underlying rationale for ALPR expansion was related to the informed belief that stolen vehicles were involved in a significant portion of gun violence in the city, we explored all fatal and non-fatal, injurious shooting incidents from January 2021 through November 2024 using data from ACPD's Violent Crimes Unit (VCU)—tasked with investigating all shootings and homicides. All shootings prior to December 2022—the month ACPD began live-monitoring the stationary/fixed ALPR alerts—were coded as a 0 and with all subsequent shooting incidents from December 2022 on coded as a 1 to account for pre- and post-ALPR expansion. A total of 135 shooting incidents occurred between January 2021 and November 2024 (fatal = 29; 21.5%); 83 incidents occurred prior to December 2022 (61.5%) with the other 52 from December 2022 through November 2024 (38.5%).

VCU detectives accessed the casefiles for each shooting and determined whether or not the ALPR information system, specifically data previously collected from documented scans and stored, was used to assist with their investigation (i.e., the system produced tangible evidence). ACPD operated a handful of mobile ALPRs mounted on police vehicles prior to November/December 2022, and VCU commonly consulted with the ALPR information system during this time period. All shootings were coded as a 0 if ALPRs were not used and assisted with the VCU investigations and a 1 if they were. Additionally, we explored whether the ALPR expansion post-December 2022 was associated with improved arrest/clearance rates for shooting incidents. VCU, however, was unable to produce such measures for 2021 at the time of this writing. Therefore, we explored arrest/clearance

rates for January 2022 through November 2024. Shooting incidents were coded as a 1 if charges were filed (i.e., cleared) and 0 if they were not.

A series of cross-tabs and Chi-square tests were conducted to examine the bivariate relationships among both ALPR use/assistance and clearance rates for such shooting incidents between time 1 (pre-ALPR expansion before December 2022) and time 2 (post-ALPR expansion). Table 3 provides the results of these bivariate analyses and it displays a statistically significant difference in the proportion of shooting incidents in which ACPD's VCU utilized the ALPR system and it assisted with their investigations post-technology expansion ($\chi^2=4.11$; $p<0.05$). While VCU did regularly use the technology and it assisted with their fatal and injurious shooting investigations before December 2022 (57.8%), this use and assistance increased to 75.0% post-ALPR expansion through November 2024. Despite a slight improvement in clearance rates from 77.1% pre-December 2022—which was already fairly high relative to most other jurisdictions (Braga, 2021) to 80.8% afterwards, the ALPR expansion was not statistically significantly associated with this increase ($\chi^2=0.17$; $p=0.68$). It is possible that the failure to include shootings from 2021 reduced the sample size and statistical power.

Similar pre- and post-ALPR expansion comparisons were performed using monthly crime data. ACPD started transitioning from the Uniform Crime Reporting (UCR) to the National Incident-Based Reporting Systems (NIBRS) during Quarters 1 and 2 of 2021. In effort to avoid data inconsistencies and/or unreliability associated with the transition, ACPD suggested that such analyses begin with July 2021. Monthly NIBRS figures—including violent crimes, property crimes, and motor vehicle thefts—were examined through November 2024 as well as the aforementioned VCU monthly totals of fatal and non-fatal shootings from January 2021 through November 2024.

Table 4 presents series of one-way ANOVAs comparing the mean monthly counts of such recorded crime incidents before and after December 2022. The F ratios

Table 3. Cross-tabs and chi-square tests for fatal and injurious shooting incidents.

Variable categories	ALPR expansion		Pearson chi-square
	Pre-	Post-	
Use/assistance of ALPRs in investigations ^a			4.11*
No	35 (42.2%)	13 (25.0%)	
Yes	48 (57.8%)	39 (75.0%)	
Clearance rates/charged ^b			0.17
No	8 (22.8%)	10 (19.2%)	
Yes	27 (77.1%)	42 (80.8%)	

* $p<0.05$ (two-tailed test).

^aJanuary 2021–November 2024.

^bJanuary 2022–November 2024.

Table 4. One-way ANOVAs.

Crime Types	ALPR expansion – monthly means		F ratio
	Pre-	Post-	
Violent crime ^a	58.4	56.6	.20
Fatal and injurious shootings ^b	3.6	2.2	7.39**
Property crime ^a	321.2	211.1	23.94***
Motor vehicle theft ^a	23.6	19.0	4.25*

* $p<0.05$; ** $p<0.01$; *** $p<0.001$ (two-tailed test).

^aJuly 2021–November 2024.

^bJanuary 2021–November 2024.

Table 5. Interrupted time-series analyses (ARIMA).

Outcome	(p, d, q)	R-squared	Coefficient	SE	T
Shootings	(0, 0, 0)	0.14	−1.33	0.27	−2.87**
Motor vehicle thefts	(1, 0, 0)	0.24	−0.49	0.15	−3.47**
Property crime	(1, 0, 0)	0.52	−0.74	0.11	−6.72***

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$ (two-tailed test).

suggest there were no statistically significant mean changes for violent crime from pre- to post-ALPR, whereas the mean monthly fatal and injurious shootings, property crimes, and motor vehicle thefts imply statistically significant reductions.

Interrupted Time Series Analyses

The one-way ANOVAs provided a cursory review of mean monthly differences in reported NIBRS offenses between pre- and post-ALPR expansion. While informative, such analyses do not account for the possibility of serially correlated errors and other issues (e.g., non-stationarity) in a time series that introduce bias. Therefore, we performed time series analyses on the data using autoregressive integrated moving average (ARIMA) models for the three crimes with preliminary evidence of reductions at time 2. ARIMA involves two step process of model identification and impact assessment. The model identification stage can be tedious with number of diagnostics: visualizing the data, plotting the value of autocorrelation function (ACF) through correlograms, and assessing the patterns of the lags to determine right p (autoregressive), d (differencing or trend), and q (moving average) components. The Expert Modeler function in SPSS finds the best-fitting model for each dependent series, including automatically transforming model variables if the time series data are non-stationary (e.g., differencing). We elected to use the Expert Modeler function in SPSS Version 29. The results from the three ARIMA models for shootings, motor vehicle thefts, and property crime—each with the inclusion of an intervention variable starting December 2022—are presented in Table 5.

In all three of the estimated models, ARIMA indicates that the ALPR expansion period was associated with a statistically significant decline in monthly shootings ($t = -2.87$; $p < 0.01$), motor vehicle thefts ($t = -3.47$; $p < 0.001$), and property crime ($t = -6.72$; $p < 0.001$) as a whole. These findings bolster the significant associations previously found in the bivariate analyses. Given the increased use of the ALPR system for shooting investigations—plus the slight uptick in their arrest rates—as well as the number of stolen vehicles recovered, arrests made and perpetrators jailed for motor vehicle thefts, the potential mechanisms at work point more toward a specific deterrent and incapacitation effect.

Discussion

Rigorously testing the impact that innovative technologies have on policing is critical. Startup and maintenance costs are expensive as departments and their respective governments must decide how available funds are dispersed between technology, personnel, and training, among other expenses. Many of these new technologies, ALPRs included,

are often aggressively marketed with the promise of increased efficiency and effectiveness without the necessary evaluation research and findings to back up their claims. This applies from acoustic gunshot detection systems (see Mares, 2022) to artificial intelligence (AI) that assists officers in writing reports (Adams et al., 2024) to ALPRs.

Our study attempted to build on the scant evaluation work for one of, if not the, most rapidly diffused technologies in American policing over the last twenty years. The aim is that it can better inform police executives and government officials of both ALPRs' costs and benefits, particularly for stationary/fixed cameras. ACPD experienced significant delays in installing their newly purchased, grant-funded cameras due to number of factors largely out of the department's immediate control. Unlike the relative ease with which ALPRs can be mounted on police vehicles, stationary/fixed cameras introduce added layers of complexity, bureaucracy, and sometimes legality that can postpone or interrupt installation. Departments cannot usually install and power these cameras themselves and are reliant on electricity/utilities companies. Perhaps more relevant are issues of ownership and agreement. Prior to installation, departments need permission from or to enter into formal, legal contracts with the authority of a particular road or piece of infrastructure on which the camera is mounted. Such issues landed one ALPR vendor, Flock Safety, into controversy as a *Forbes'* report detailed how the company failed to obtain correct permitting for hundreds of cameras installed on state infrastructure without proper approval from state departments of transportation in South Carolina, Texas, Washington, and Illinois (Brewster & Farivar, 2024).

Aside from impediments to installation, ACPD's stationary/fixed cameras regularly become inoperable and require a certain degree of maintenance for repowering, resetting, and adjusting angles for ideal scanning. The ALPR system captured only 61.0% of the total camera-month hours between November 2022 through November 2024 with the inbound cameras, which provide for more actionable activity, performing more poorly in terms of active hours scanning compared to the outbound cameras. Some of the vague (e.g., "other") and less serious alert types (e.g., "suspended or revoked") occur continuously with the overwhelming majority of these not being monitored or addressed.

In fact, it appears ACPD officers, particularly patrol, are stretched too thin and perform a trivial amount of vehicle stops on the more serious ALPR alerts, such as "stolen vehicle" or "wanted" persons. Over a twenty-five month period, ACPD documented only 72 initiated vehicle stops following ALPR alerts resulting in 40 stolen vehicles and 9 stolen license plates recovered as well as 66 of those occasions when charges were filed and people were taken into custody. One ALPR alert inbound for a stolen vehicle in August 2023, in particular, led to a vehicle stop that yielded charges of unlawful possession of a handgun, which was defaced, as well as possession of hollow point ammunition and a high-capacity magazine. Another 18 stolen vehicles and 5 stolen license plates were recovered with the assistance of the ALPR technology after those vehicles were found abandoned/unoccupied (58 stolen vehicles and 14 stolen licenses plates recovered in total); ACPD also relayed 59 notifications regarding outbound ALPR alerts to neighboring police departments.

While commendable, it is unclear how much more of an impact the technology expansion could have had on patrol-related functions if officers were able to respond to a higher proportion of active ALPR alerts. Retired officers now acting in a civilian capacity

in ACPD's 24/7 surveillance center are also partially to blame, as they should have been calling out and dispatching more patrol officers to ALPR hits/alerts in real time. A larger commitment to such a response, however, would certainly result in commensurate increases in workloads for officers, which may negatively affect ACPD's ability to respond to traditional calls for service or engage in other types of proactive activity (e.g., foot/bike patrol; hot spots policing)—much like those found in St. Louis, MO following an expansion of the city's AGDS (Blackburn & Mares, 2019; Mares & Blackburn, 2012). It would likely create and illuminate another hidden cost of the ALPR technology.

There is more evidence that the ALPR expansion impacted investigation-related functions compared patrol-related functions. From December 2022, ACPD's VCU detectives used data from previous ALPR scans and its information system to assist with fatal and non-fatal shootings in nearly three-quarters of such incidents through August of 2024. Clearance rates for these crimes improved over this time period, although they were not statistically significant—similar to the findings from Koper and Lum's (2019) evaluation in Charlotte, NC. Use of the ALPR system seems to be more institutionalized in some specialty units within ACPD, such as VCU. Aside from shootings investigations, the ALPR system was also provided crucial evidence in the arrest of a person who committed a fatal hit and run of a pedestrian in the city in March 2023 plus another pair of hit and runs in 2024. Future research should continue to examine how ALPRs and their information systems contribute to investigative and intelligence-related efforts since the extant literature is so deficient in this regard.

There is little evidence that the ALPR expansion was associated with reductions in violent crime in general. However, the results from both the one-way ANOVAs and ARIMA models suggest that the changes post-December 2022 were associated with decreases in shootings, motor vehicle thefts, and property crime in general. These findings are promising; however, it should also be mentioned that it is difficult to rule out other factors—both locally and nationally—that might be contributing to those reductions, particularly during similar time intervals as the ALPR expansion. ACPD engages in a number of data-driven and evidence-based policing techniques, including risk terrain modeling (RTM). ACPD and the City partnered with the New Jersey Department of Community Affairs (DCA) and Stockton University to introduce CitiStat and CompStat programs to address quality of life and public safety issues with their first bi-weekly meeting taking place in mid-November 2022 (Atlantic City, 2022; DCA, 2022). Additionally, most crime trends in cities across the US started to decline in 2023 from their recent peaks during 2020–2022, specifically gun violence (Asher, 2024) but not motor vehicle thefts (Council on Criminal Justice, 2024). The challenging task from a research standpoint is trying to disentangle the effects from local efforts and broader national trends.

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